

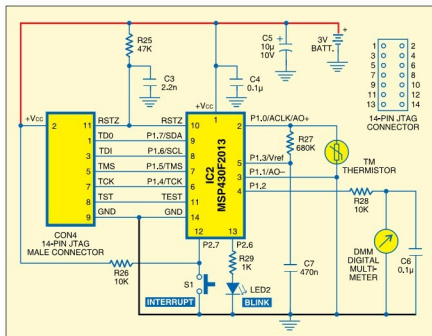


Fig. 10: Circuit of blinking LED cum digital thermometer

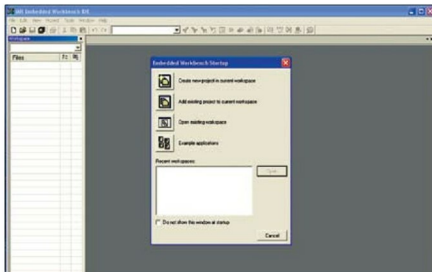


Fig. 11: Screenshot of 'Create New Project' in IAR Embedded Workbench

MSP430 target board to the 14-pin FRC connector from the ParaFET board.

3. Start IAR system from the desktop.
4. Select 'Create New Project' option from 'Project' menu (refer Fig. 11).
5. Give a suitable project name, e.g., BlinkingLED.
6. Select 'New File' from 'File' menu. It will open up an empty document.
7. Copy and paste the blinking LED

program to the empty document.

8. Right-click 'Untitled' tab and enter the file name with '.c' extension, e.g., `blinking_led.c`
9. In the left window pane, right-click the project name and select 'Add' option to include 'blinking_led.c' program.
10. Right-click the project name and select 'Options' (refer Fig. 12).
11. Left-click the square button next to the text box containing 'Ge-

neric MSP430 Device' and select 'MSP430F2013.'

12. Click 'Library Configuration' and select 'Normal DLIB' for library option.
13. Click 'Library Options' tab and select 'Tiny' for printf formatter and 'Small' for scanf formatter; these settings ensure small size of the compiled 'C' code, which is appropriate for the applications given in this article.
14. Click 'ok.'
15. Again, right-click the project name and select 'Debugger' from 'Options' pull-down menu. Here, you can choose between simulator and FET debugger options.

16. Select 'Simulator' to simulate the program in software and 'FET Debugger' to download the program using ParaFET to the target board in order to program the MCU and do an in-circuit emulation. Ensure that the FET debugger is either Texas Instruments LPT-if or just parallel-port-if.

17. Invoke the simulator/debugger by clicking the icon shown in Fig. 13.

18. On clicking the debug button, the IAR will compile the program and give a message saying that it is erasing the flash and downloading the program. If the program download is successful, the debugger environment comes up. Now, the program is ready to run on the target board.

19. Remove the ParaFET circuit and test the blinking of the LED. The LED should blink with 'on' and 'off' durations of about one second each. The operation steps are detailed in this month's EFY-CD.

Tips

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